Typed Epistemic State under Partial Knowledge: Matched-Information Mechanism Studies and Real-Data Boundaries

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Abstract

Research systems act on more than world facts: they also carry partial knowledge about feasibility, failed interfaces, uncertain costs, transported certificates, unresolved questions, and representation edits. We ask when explicitly typing and scoping that epistemic state changes a decision while competing mechanisms receive the same visible information. Six prospectively frozen exact-synthetic studies isolate one operation at a time. Type-conditioned value-of-information planning exceeds an otherwise identical uniform-prior planner; scope-bound reopening decisively beats reopening on unscoped change, although its advantage over never reopening is not separated from zero in either regime. Matched-budget targeted verification lowers scalarized regret, full-chain checking detects every constructed laundering chain, decision-coupled acquisition avoids irrelevant high-entropy probes, and typed remint/transport improves utility only when transport is useful, tying re-derivation exactly in a prespecified no-value regime. A corrected finite decision criterion explains the last pattern: refinement lowers Bayes risk strictly exactly when no action is simultaneously optimal across all refined subfibres of at least one coarse fibre. This is donor decision theory, not a theorem-novelty claim. We instantiate the criterion on five OpenML benchmark datasets and twelve Defects4J projects with both value and no-value cases; same-distribution agreement is algebraic, not independent confirmation. Held-out results are mixed: the OpenML binding captures a negative fraction of the oracle gap on three of five datasets, while the Defects4J refinement helps on ten of twelve projects but has one genuine unexplained failure. A WorkflowHub study lowers held-out regret on one corpus yet cannot test the no-value direction because every stratum predicts benefit. The evidence supports controlled mechanism isolation and bounded real-data transfer, not deployed-agent, security, universal-necessity, or practical-superiority claims.

Keywords: epistemic state; partial knowledge; value of information; provenance; verification; active experimentation.

1 Introduction

A long-running research system accumulates facts about its own knowledge state: an interface is known feasible, unknown, or previously failed under a particular representation; a cost is measured or only bounded; a certificate was minted under an earlier context; an unanswered question is uncertain but may or may not affect the current decision. Flattening these states into generic booleans or confidence scores can erase information about when evidence remains applicable.

The scientific difficulty is attribution. If a typed system has more information than an untyped baseline, a positive result says little about typing itself. This study therefore uses matched-information worlds. Inside each family, every non-oracle arm receives the same serialized visible state and differs only in how it interprets or acts on that state. The strongest comparison gets first right of refusal, and each world contains a hostile regime designed to expose the shortcut that the proposed typing is meant to prevent.

Six synthetic families cover complementary operations: priors over unknown feasibility, scope-bound invalidation of failure receipts, verification under interval uncertainty, provenance-chain transport, decision-coupled experiment selection, and remint/transport across representation changes. Their purpose is not realism. Exact construction lets the mechanism be isolated against enumerable truth, matched information, and paired outcomes.

The controlled studies leave two questions. First, what finite condition determines whether a refinement can improve the best decision rule? Second, does the same idea remain useful when the fibres are constructed from real records and actions are estimated from a development split? We answer the first with a corrected decision-theoretic criterion and address the second with bounded studies on OpenML benchmark datasets, Defects4J, and WorkflowHub records. These real-data studies are deliberately interpreted through their held-out results rather than through same-distribution identities.

Negative evidence is part of the design rather than an appendix. One synthetic family loses its policy residual to an ideal value-of-information donor, and another candidate is donor-absorbed on its original world. On real data, held-out transfer is mixed, one post-hoc explanation fails, and the WorkflowHub corpus cannot test the no-value half of the criterion. A mechanism-isolation programme that reports only its wins would obscure the conditions under which its proposed state representation adds nothing.

2 Methods

2.1 Prospective freezing and paired construction

Each study freezes its world generator, seed, arms, metrics, gates, tie-breaks, and honest terminal vocabulary before result-bearing execution. Episodes are paired where relevant so competing arms face the same underlying realization.

2.2 Matched information

Within a family, every non-oracle arm receives the identical serialized visible state. Mechanism comparisons therefore change the decision rule, not the available facts. Oracle arms are included only as upper/reference bounds and are not described as deployable baselines.

2.3 Finite decision criterion

Let X be a finite set of worlds with mass $p(x) > 0$, let A be a finite action set, and let $\ell(x, a)$ be loss. A binding B partitions X into fibres F. The minimum risk attainable by a deterministic B-measurable policy is

$$R(B) = \sum_{F \in B} \min_{a \in A} \sum_{x \in F} p(x)\ell(x, a).$$

A zero-oracle-regret B-measurable policy exists exactly when every fibre has an action that is optimal in every world in that fibre. If B' refines B, then $R(B') \le R(B)$. For a B'-subfibre G, define O_G as the actions minimizing aggregate loss on G. The inequality is strict exactly when at least one coarse fibre has $\bigcap_{G \subseteq F} O_G = \varnothing$ across its positive-mass refined subfibres. If the intersection is nonempty, one action attains every subfibre minimum and the risks agree on that fibre; if it is empty, no coarse action can attain their summed minima. This is a finite decision-sufficiency identity rather than a claimed new theorem. An earlier shorthand that referred only to splitting an “impure fibre” was too broad and is withdrawn.

2.4 First right of refusal

The strongest donor-complete comparator appropriate to each construction is registered in advance. A donor tie or win closes the mechanism or policy claim. This rule is responsible for the costly-verification donor and original-world model-selection donor negatives reported below.

2.5 Hostile validity gates

Worlds include a regime that should punish the shortcut under test: blind optimism, global reopening, partial chain inspection, decision-irrelevant information gain, stale carry-forward, or a typed operation that should provide no value. The expected trap must actually bite where prespecified; otherwise the world terminates invalid rather than yielding a positive mechanism claim.

2.6 Determinism and authority

The synthetic studies use frozen seeds and deterministic decision paths. Canonical receipts encode result and scope. Arms labelled as language-model proxies refer to fixed heuristics, not to measurements of any deployed model. The result boundary is exact-synthetic mechanism isolation.

2.7 Real-data instantiations and held-out evaluation

We used three public data sources with frozen actions and losses. The OpenML benchmark suite CC18 supplies binary classification datasets and standardized metadata Bischl et al. (2021); Vanschoren et al. (2014). For each of five datasets, quantile-binned coarse and refined bindings were fitted on one half; the same half determined the optimal fibre actions and the finite-criterion instantiation, while the other half measured transfer. Defects4J supplies modified classes and tests associated with 597 faults in twelve Java projects Just et al. (2014). Candidate test classes were scored under a fixed run/skip utility. A project was the scientific unit, and held-out bugs were not used to choose fibre actions. WorkflowHub supplied 1,533 registry records describing computational workflows Gustafsson et al. (2025); the decision was whether to attempt parsing an RO-Crate, using licence as the coarse binding and workflow class as its refinement Soiland-Reyes et al. (2022).

The same-distribution criterion check is algebraic: once subfibre-optimal actions and risks are computed from the same empirical distribution, agreement is a consequence of the identity above. We therefore treat it as an implementation and domain-instantiation check, not independent confirmation. Scientific transfer is assessed only on the held-out halves.

2.8 Deviations and analysis status

The OpenML analysis was corrected after initial outcomes exposed two specification errors: the first version scored the finite identity out of sample, and a minimum-mass threshold was applied to the prediction but not to the risk calculation. Because those repairs followed outcome inspection, the corrected OpenML results are descriptive rather than confirmatory. For Defects4J, a proposed binding produced 72.7% singleton fibres; it was replaced after inspecting fibre sizes but before catch outcomes were computed, and a 50% degeneracy gate was added. The WorkflowHub protocol initially stratified by the same feature introduced by the refinement, making the within-stratum refinement an identity. The stratifier was changed after that outcome, but all replacement strata still predicted value. The study therefore retains a cannot-assess conclusion for the required two-sided contrast rather than searching for a favorable stratum.

Synthetic paired intervals are percentile-bootstrap intervals over the registered paired episodes. They are reported per comparison without family-wise correction. The real-data project and dataset counts are descriptive; rows, tests, bugs, and records are not treated as independent replications of cross-domain transfer.

3 Results

3.1 Typed priors for unknown feasibility

Type-conditioned priors are the only difference between the typed value-of-information arm and its uniform-prior ablation. Over 300 paired episodes, the typed arm has mean utility 3.291 versus 2.180 for uniform value of information and 4.612 for the full oracle, recovering about 71% of oracle utility. Blind optimistic commitment is driven to -13.619 mean utility and succeeds in only 19% of episodes, satisfying the hostile-world validity gate.

3.2 Scope-bound failure-receipt reopening

The reopening family separates changes that invalidate a failure receipt from irrelevant context churn. Pooled mean utility is 3.199 for scoped reopening, 2.782 for never reopening, -7.813 for unscoped-change reopening, and -9.225 for always reopening. In the deliberately wasteful regime, always reopening collapses to about -13.406 while never reopening remains positive; the scoped mechanism retains the no-giveback property required by the protocol.

The scoped-versus-never gap is not a supported result, and we state this explicitly because the pooled means invite the opposite reading. Under the paired analysis of Table 1, scoped reopening is not separated from never reopening in either regime: the paired mean difference is 0.060 with 95 % interval $[-0.540, 0.634]$ in the wasteful regime and 0.774 with interval $[-0.663, 2.254]$ in the stale regime, on 200 pairs each. Both intervals contain zero. The wasteful-regime comparison is additionally 79 % ties, so the pooled means are separated mostly by a small minority of discordant episodes. What the family does establish at this sample size is the scoped-versus-*unscoped* contrast, where both intervals exclude zero by a wide margin ($[13.624, 16.451]$ and $[5.740, 8.255]$). The mechanism claim is therefore that scoping reopening beats reopening on unscoped change — not that reopening under scope beats never reopening at all.

3.3 Targeted verification under interval costs

Every active arm in the verification family receives the same budget of four edge verifications. Interval-dominance targeting yields mean scalarized regret 0.1096, compared with 0.2518 for random verification, 0.2621 for midpoint optimization without targeted verification, 0.7755 for worst-case optimization, and 1.2679 for best-case optimization. The benefit is therefore attributed to where verification is spent, not verification volume.

3.4 Full-chain certificate transport

The transport census contains 200 honest and 200 laundering chains. The full-chain checker has recall 1.000 and false-positive rate 0.000. It detects all missing-receipt, spoofed-summary, and 68 deep-splice attacks. Last-hop inspection has zero recall on the deep-splice class. This is a construction-level provenance result; the identifiers are seeded values rather than cryptographic digests.

3.5 Decision-coupled active experiments

In the acquisition family, pure information gain is attracted by high-entropy facts that are irrelevant to the downstream choice. It spends 36.6% of its probes on decoys and obtains mean utility 7.121, while the decision-coupled selector spends none on decoys and reaches 9.266 under the shared stopping rule. The decoy-attraction gate is part of world validity.

3.6 Remint and transport across edits

The remint/transport family compares typed transport with naive carry-forward and matched-budget re-derivation. In the stale-hostile regime, naive carry-forward fails on 99.5% of episodes and has mean utility about -15.916; typed transport reaches 9.421 versus 7.157 for re-derivation. In the prespecified remint-unnecessary regime, all relevant arms tie exactly at 11.809659685355605 and the typed arm spends zero remints. A positive advantage there would have invalidated the study.

Table 1: Paired uncertainty for every synthetic comparison in the preserved publication analysis. Intervals are 95 % percentile bootstrap intervals over paired episode differences (5000 deterministic resamples, seeds recorded per row in the source artifact). Win/tie/loss is the paired decomposition over the same n pairs. Intervals are per-comparison and are *not* adjusted for the twelve comparisons shown; any family-wise adjustment widens them, so rows already containing zero would continue to contain zero. Rows marked † have an interval containing zero: the contrast is *undetermined* at this n and is not a positive result. The row marked ‡ is the prespecified remint-unnecessary regime, where the exact tie on all pairs is the passing outcome and a positive advantage would have invalidated the study; its interval is $[0, 0]$ by construction and is not an undetermined contrast.

Study	Comparison	Mean diff.	95 % CI	n	win/tie/loss
Prior	typed VOI vs. greedy known-graph	2.933	[2.556, 3.301]	300	0.55/0.30/0.15
Prior	typed VOI vs. uniform-prior VOI	1.111	[0.833, 1.400]	300	0.29/0.58/0.13
Reopening	scoped vs. never reopen (wasteful)†	0.060	[−0.540, 0.634]	200	0.17/0.79/0.04
Reopening	scoped vs. unscoped reopen (wasteful)	15.050	[13.624, 16.451]	200	0.78/0.11/0.11
Reopening	scoped vs. never reopen (stale)†	0.774	[−0.663, 2.254]	200	0.59/0.20/0.21
Reopening	scoped vs. unscoped reopen (stale)	6.973	[5.740, 8.255]	200	0.52/0.35/0.13
Verify	interval-Pareto vs. random verification	0.142	[0.100, 0.187]	400	0.26/0.67/0.07
Acquire	decision VOI vs. information gain	2.146	[1.976, 2.299]	400	0.95/0.03/0.02
Acquire	decision VOI vs. LLM proxy	0.277	[0.119, 0.412]	400	0.45/0.44/0.14
Transport	typed transport vs. naive carry-forward	17.242	[15.577, 18.836]	200	0.69/0.30/0.00
Transport	typed transport vs. re-derivation (mixed)	2.264	[1.717, 2.825]	200	0.32/0.69/0.00
Transport	typed transport vs. re-derivation (no remint needed)‡	0.000	[0.000, 0.000]	200	0.00/1.00/0.00

Metric note: Prior, acquisition and transport differences are in episode utility; reopening in mean round utility; verification in scalarized regret reduction. The provenance family is a frozen exact census (200 laundering / 200 honest chains), not a sampled paired comparison, and therefore has no interval; its counts are reported in the text.

3.7 Paired uncertainty for every comparison

The means reported above are point estimates. Table 1 gives the paired 95 % percentile bootstrap interval, the pair count, and the win/tie/loss decomposition for each of the twelve synthetic comparisons. Three readings matter for how far these results should be taken.

First, two comparisons are undetermined: both scoped-versus-never reopening contrasts have intervals containing zero, as discussed above. Every other non-degenerate comparison excludes zero.

Second, several intervals that exclude zero rest on a minority of discordant pairs, and the tie column shows how often the mechanism changes anything at all. Typed-versus-uniform value of information wins 29 % of pairs against 13 % losses with 58 % ties; targeted verification wins 26 % against 7 % with 67 % ties. Both differences are consistent in direction, but they are carried by roughly a third of episodes, not by a uniform shift. The large-margin results — decision-coupled acquisition versus information gain (95 % wins) and typed transport versus naive carry-forward (69 % wins, zero losses) — are the ones that hold on most episodes.

Third, the intervals are per-comparison and unadjusted for multiplicity. Twelve comparisons are reported, nine of which exclude zero. Any family-wise adjustment widens all of them. The one direction that follows without further computation is that the two undetermined contrasts can only remain undetermined. We do not assert here which of the nine would survive adjustment. The preserved analysis stores summary statistics rather than the paired difference vectors, so corrected *bootstrap intervals on the mean* cannot be derived from it. A family-wise-correctable exact test is available by a different route, over the win/loss counts rather than the means, and is reported separately; we keep the unadjusted intervals here because the mean is the registered estimand for these comparisons. Targeted verification and decision-coupled acquisition against the deterministic proxy are the comparisons most exposed to a correction because their intervals lie closest to zero relative to their width.

3.8 Negative first-refusal cases

A separate failure-state study finds a typed/scoped benefit over a scoping ablation, but an ideal value-of-information allocator given the same typed facts matches the candidate policy exactly at solve rate 0.9866. The allocation-policy residual is therefore closed. A model-selection study similarly gives its donor first refusal: candidate and donor both score 0.9948 on the original world, leaving only the narrower misspecified-world edge (0.9844 versus 0.9531) as a surviving bounded result.

3.9 Real-data instantiations do not establish the criterion

On the five OpenML benchmark datasets, four development distributions contain refined subfibres with incompatible optimal actions and one does not; the computed risk changes in the corresponding direction in all five cases. Across the twelve Defects4J projects, eleven development distributions have the value condition and one does not, again with matching same-distribution risk changes. These counts exercise the implementation on real records and populate both sides of the criterion, but they are not independent tests of an identity computed from the same empirical losses.

The held-out results answer a harder question and are mixed. On OpenML, the typed refinement captures a negative fraction of the oracle-achievable gap on three of five datasets and is beaten by generic information-gain refinement on three. On Defects4J, refinement lowers held-out regret in ten of twelve projects. One exception is the no-value project, where there is no development gain to transfer. The other, Cli, has a genuine development gain but worse held-out regret (0.0256 for the coarse binding versus 0.0278 after refinement). Candidate-set size, development sample size, fibre coverage, and in-sample gap captured do not distinguish Cli from successful neighboring projects.

In 1,533 WorkflowHub records, adding workflow class to licence lowers regret from 0.1031 to 0.0065 on the development distribution and from 0.1076 to 0.0137 on held-out records. However, every one of four tag-count strata predicts value. The required no-value stratum is absent, so this corpus cannot test the two-sided criterion. The positive transfer measurement stands as a single-corpus result; it is not a third confirmation.

4 Discussion

Across the six primary families, the common pattern is conditional rather than universal. Typing/scoping matters when it changes which uncertainty, receipt, certificate, or experiment is relevant to the decision. The matched-information ablations make that interpretation stronger than a comparison in which the typed arm simply sees richer state.

Equally important is the no-value behavior. The exact remint/transport tie shows that the mechanism does not need to win everywhere to support its intended claim; where transport has no decision value, the typed mechanism does nothing and ties. The corrected finite criterion makes this precise at the level of aggregate subfibre decisions, without turning generic decision sufficiency into a novelty claim. The ideal-donor tie pushes the same principle further: typed state can matter while a proposed policy adds nothing beyond a complete comparator. The method claim should stop at the representation in that case.

Hostile controls are useful because they test whether a synthetic world actually exercises the proposed failure mode. They do not make the world realistic. A model can perform well on all six constructions and still fail on real interfaces for reasons absent from the generators: mis-specified state types, noisy provenance, nonstationary costs, strategic adversaries, or poor semantic extraction from raw evidence.

The real-data results show why same-distribution mechanism value and transfer must remain separate. The finite criterion correctly characterizes the empirical distribution on which fibre actions are chosen, but it says nothing about estimation error after refinement creates more, smaller fibres. OpenML exposes this sharply: the typed refinement is exact in-sample and harmful on three held-out datasets relative to the attainable gap. Defects4J is more favorable, yet one project with genuine development value still worsens out of sample and no recorded project-level statistic explains why. WorkflowHub adds a strong held-out improvement but lacks the no-value contrast needed to probe the full criterion.

The bounded conclusion is therefore not that typed state is generally useful. It is that type and scope can be isolated as decision variables under matched information, that a finite optimal-action criterion states when refinement has value on a fixed distribution, and that transfer must be measured rather than inferred from either exact synthetic success or same-distribution identities. A stronger practical claim would require prospectively frozen, independently governed domains with enough independent units to estimate when refined fibres generalize.

5 Related work and boundary

The component mechanisms have established lineages. Robust optimization studies decision making under uncertain inputs and the trade-off between nominal performance and protection against uncertainty Bertsimas & Sim (2004). Data-provenance work formalizes where derived records originate and which source data influence them Buneman et al. (2001), while active-learning literature studies how to choose information-gathering actions rather than passively consume labels Settles (2009). None of these ingredients is claimed individually.

Recent agent work makes the boundary substantially tighter than those classical foundations alone. STALE directly benchmarks whether agents recognize and act on invalidated memory Chao et al. (2026). MAP-Graph uses a typed provenance graph, ancestry, permission filtering, path trust, and action gating as an operational memory control Wang et al. (2026), and provenance-sensitivity auditing varies source authority while holding other task factors fixed Liao (2026). Long-horizon studies also report loss of governance constraints under context compaction Chen (2026) and different decay patterns for omission and commission constraints Gamage (2026). We therefore do not claim priority on typed memory, provenance-sensitive action, stale-state revision, or governed/versioned memory.

The closest recent study examines budgeted verification of inherited stale constraints Nakayashiki (2026). It fixes the verification budget at two records in every arm and includes a random-record control, so neither matched-budget verification nor a random-verification control is a novelty claim here. That work measures a sampled behavioral effect under fixed allocation. The residual object in this paper is narrower: controlled mechanism isolation under matched visible information, together with a finite criterion for whether refined subfibres require different optimal actions. The generic criterion belongs to decision-sufficiency and Blackwell-style comparison theory; the present contribution is its bounded operationalization across the six mechanism families and the accompanying transfer boundaries.

The exact-versus-sampled distinction should not be read as an external-validity claim. The budgeted-verification study reports large-effect and near-null conditions consistent with the finite criterion, but no episode-level fibre mapping is performed here. More importantly, the present OpenML and Defects4J instantiations show that same-distribution consistency does not imply held-out benefit.

The bounded contribution is therefore a common mechanism contract across six separately frozen exact-synthetic partial-knowledge operations: hold the factual world fixed, vary the explicit epistemic binding available to the decision rule, use a matched comparator, include hostile or no-value controls, and report both value and no-value outcomes. The cross-family taxonomy is a post-study synthesis. The real-data studies add bounded transfer evidence but do not establish deployment prevalence or superiority.

Ignorance is already known not to be one thing. The premise that epistemic state needs typing rather than a single not-known flag has independent formal support. Kubyshkina and Petrolo give sound and complete systems for three distinct notions — ignorance whether, ignorance as unknown truth, and disbelieving ignorance, in which the agent holds a false belief about a true proposition Kubyshkina & Petrolo (2026). The distinctions are load-bearing rather than terminological: disbelieving ignorance is not interdefinable with knowledge on the relevant frames, so a system that records only *not known* cannot express it. Their topic-sensitive semantics adds a further separation, since an agent may be ignorant of one of two logically equivalent propositions when she does not grasp the topic of the other.

That literature settles the conceptual question this paper would otherwise have to argue, and settles it more rigorously than an empirical study could: the types are formally distinct, and collapsing them loses expressive power. What it does not address is whether the distinction is *observable* in a running system. An axiomatization tells you the states are different in principle; it does not tell you whether two mechanisms that are supposed to occupy different epistemic states behave differently under matched information, or whether an implementation that claims to distinguish them actually does. That is the question the mechanism studies here are built to answer, and it is why the information available to each arm is matched rather than merely reported.

6 Limitations

The six primary mechanism studies are exact synthetic. Their enumerable truth and cleanly typed state provide internal control but omit the extraction and misclassification errors faced by real systems. The three real-data studies construct bindings from existing metadata rather than observing a deployed research agent, so they do not close that external-validity gap.

The language-model-labeled baselines are deterministic proxies. No conclusion about current or future language models follows. The certificate-transport family does not study cryptographic attackers; its hashes are construction identifiers and its completeness statement assumes end-to-end receipts cannot be forged consistently within the world definition.

The verification family reports scalarized regret, not full Pareto-front quality. The reopening family excludes intra-episode receipt accrual, and two of its comparisons are undetermined rather than positive: scoped versus never reopening has a paired interval containing zero in both regimes (Table 1), so the study is not entitled to a scoped-versus-never claim in either direction. The programme has no lower-bound or impossibility theorem showing that typing is necessary for every mechanism family.

Reported intervals are per-comparison. Twelve synthetic comparisons are reported without a family-wise correction. The preserved publication analysis stores summary statistics rather than paired difference vectors, so corrected bootstrap intervals on the mean cannot be recomputed from it; an exact test over win/loss counts is correctable but addresses a different estimand and does not license a claim the mean intervals fail to support. Targeted verification and decision-coupled acquisition against the deterministic proxy are most exposed because their intervals are closest to zero relative to their width.

The real-data evidence is also limited. OpenML was corrected after outcome inspection and is descriptive. The Defects4J binding was amended after fibre-size inspection, although before catch outcomes, and the project count is twelve. WorkflowHub contributes one corpus and retains a cannot-assess conclusion for the missing no-value contrast. Neither rows nor bugs are independent cross-domain replications. Multiple positive synthetic families likewise do not constitute independent populations, and no prevalence claim follows.

Finally, the earlier shorthand for refinement strictness was too broad and has been withdrawn. The corrected theorem concerns simultaneous optimality across refined subfibres on a fixed distribution. It does not say that a refinement transfers out of sample, remains useful under drift, or supplies a minimal or complete epistemic-state schema.

7 Conclusion

On six frozen partial-knowledge constructions, explicitly typed and scoped epistemic state changes decision quality even when compared mechanisms receive the same visible information. The effect appears in priors, receipt invalidation, verification targeting, provenance-chain checking, experiment choice, and remint/transport. Prespecified hostile controls show that the corresponding shortcuts are exercised, while an exact no-value tie and two donor-closure cases bound the claim.

A corrected finite criterion identifies when refinement lowers risk on a fixed distribution, but the real-data studies show that this identity does not guarantee held-out value. The result is therefore a mechanism-isolation benchmark with bounded transfer evidence, not a deployment result. It supports studying epistemic-state type and scope as first-class decision variables while requiring practical benefit to be established separately.

8 Reproducibility

The release artifact contains the synthetic protocols, generators, result records, paired-analysis transcription, corrected finite-theory checker, and the three real-data study scripts and frozen outputs. Each synthetic family uses a frozen seed, explicit validity gates, and terminal semantics that distinguish mechanism failure from an invalid world. The real-data records retain every amendment, post-hoc analysis, held-out failure, and cannot-assess outcome.

A reproduction should first verify the protocol/result hashes, then rerun each world and registered arm, recompute the paired comparisons and hostile validity gates, and execute the corrected theorem checker without importing the synthetic generators. The two donor-absorbed studies must be verified with the same prominence as positive families. Real-data reruns require the pinned public-source versions; the release records exact upstream identities and preserves fetched metadata where redistribution is permitted.

9 Ethics, safety and resource statement

No human participants, animals, clinical data, or personal data are used. Because the studies construct hostile failure modes, the principal reporting risk is over-interpreting those constructions as security or deployment evidence. The manuscript therefore labels them as exact-synthetic mechanism tests and keeps cryptographic, real-adversary, and real-agent claims outside the authority boundary.

Computational work is deterministic and bounded by the frozen episode counts/enumerations. Negative and donor-absorbed results are retained rather than discarded. No result here can self-authorize a promotion to a broader scientific claim, a novelty claim, a deployment decision, or any physical quantum claim.

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